

ODUGUWA DAMILOLA JOHN

◇ Lagos, Nigeria

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Software Engineer specializing in building production AI/ML Systems and scalable backend infrastructure. Experience building end-to-end ML pipelines, model training and inference infrastructure, and LLM-powered applications.

EXPERIENCE

Machine Learning Engineer

[Bluechip Technologies](#)

October 2025 - Present

Lagos, Nigeria

- Architected and deployed a Finite State Machine-based document routing system with dynamic RBAC for NLNG's IVCO scheduling platform, automating critical approval workflows between global shareholders and buyers. Reduced manual schedule iteration times by 80% and streamlined communication with international stakeholders through structured email/notification systems
- Co-designed and implemented a constraint optimization engine using Google OR-Tools (CP-SAT) to generate annual and monthly cargo delivery schedules for Nigeria's largest natural gas provider. The system accounts for tidal windows, berthing constraints, supply shutdowns, and maintenance schedules, reducing schedule generation times by 60% and enabling rapid adaptation to operational changes.
- Built a scalable event-driven backend using Azure Service Bus, asyncio, and Redis, enabling reliable asynchronous notifications at scale while improving API performance through effective caching and cache invalidation strategies.
- Developed LLM-powered schedule insights feature using agentic workflows and structured outputs, providing scheduling officers with AI-driven recommendations for adjusting schedules based on operational incidents and constraints.
- Significantly reduced cargo delivery schedule generation times by 60 %, enabling NLNG to adapt to buyer changes and operational constraints rapidly.

Machine Learning Engineer(Contract)

[Hobwise](#)

July 2025 - October 2025

Lagos, Nigeria

- Built production ML infrastructure from scratch for hospitality analytics platform, developing end-to-end pipelines using Apache Airflow and Spark for demand forecasting and customer predictions serving restaurants and hotels.
- Deployed scalable ML serving infrastructure on GCP using Terraform IaC, provisioning Vertex AI, Cloud Run, and Pub/Sub resources with automated CI/CD pipelines across dev, staging, and production environments.
- Containerized MLflow, model training pipelines, and FastAPI services; integrated them with **Kubeflow** to orchestrate distributed workloads and streamline experiment tracking and deployment.

Software Engineering Intern

[Stanbic IBTC](#)

April 2025 - October 2025

Lagos, Nigeria

- Contributed to the development of a Face Verification service using Azure Cognitive Services for BVN-based identity verification across internal platforms.
- Built an encryption API that mirrors the bank's production GCM-based encryption system, enabling the middleware team to securely encrypt and decrypt requests during local development and testing.
- Authored comprehensive technical documentation, release notes and changelogs for backend microservices, reducing new team member onboarding time by 25% and improving knowledge sharing across engineering teams

Machine Learning Engineer

[AIplier](#)

August 2024 - Jan 2025

Louisiana, USA

- Designed and implemented LLM-based Computer Use agent using Playwright and LangGraph to automate job applications across multiple job boards.

- Built a visual annotation system using bounding boxes to enable LLM navigation of web elements, and developed DOM parsing scripts to structure interactive elements into hierarchical trees for LLM consumption.

SKILLS

- **Languages:** Python, C#.
- **ML/AI** PyTorch, Triton, Langgraph, Mlflow, Apache Airflow
- **Backend** FastAPI, Redis, PostgreSQL, MongoDB, Celery, WebSockets, Microservices.
- **Infra** Docker, Terraform, GCP, AWS, Azure, CI/CD
- **Tools:** Git, Bash, Shell Scripting

EDUCATION

Bsc. Computer Engineering, University of Lagos

2020 - 2026

PROJECTS

SabiYarn.Pretrained a language model entirely from scratch on 11 major Nigerian Languages.

- Implemented loss masking strategies to optimize compute efficiency during pretraining.
- Wrote distributed training and evaluation scripts for pretraining on multi-GPU, single Node compute unit, implementing optimization schemes like NUMA-Aware CPU and memory pinning, Transparent Huge Pages and Custom Triton kernels.
- Designed and tested alternate architectures including differential attention, cut-cross entropy, and multi-token prediction, to evaluate impact on pretraining performance.

Zeno A Natural Language-based book recommendation app for discovering books based on Natural Language text descriptions built using FastAPI, AWS and Langgraph. [Try it out here](#)

Whatsol A Solana-wallet backend for sending and receiving SOLs from WhatsApp contacts. [Try it out here](#)

AIpplierAn LLM-based web automation agent for automating job applications on popular job boards

RESEARCH PUBLICATIONS

- [SabiYarn: Advancing Low Resource Languages with Multitask NLP](#) Pretraining. Published at [ACL 2025](#).